

Electrical Loads:

$$P = \frac{W}{t} = \frac{\text{energy}}{\text{time}} \quad \left. \vphantom{P = \frac{W}{t}} \right\} \text{Remember that potential energy is defined as } U = Uq$$

$$\text{So: } \frac{\text{energy}}{\text{time}} = \frac{\text{voltage} \cdot \text{charge}}{\text{time}},$$

$$\text{current, } I = \frac{q}{t}, \text{ so: } P = V \cdot I$$

for the spindle: $V = 120V$ from wall
 $P = 500W$, at peak capacity

$$\text{so } I = 4.167 \text{ A}$$

for 2 nema 17 motors: $V = 36 \text{ V}$

wire must
be rated

$$P = ?$$

$$\rightarrow I = 3.0 \text{ A}$$

$$\rightarrow P = 108W \rightarrow P = 108W, V = 120V$$

$$\text{so } I = 0.9 \text{ A}$$

wire must
be rated

for nema 23 motor: $V = 36V$ $I = 4.0A$

$$P = 144W \quad I = 1.2A \quad \therefore \text{At most, the fuse must be rated well$$

$$\text{above } \Sigma I = \boxed{5.367 \text{ A}} \quad \text{for safety}$$